

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF SCIENCE

FACULTY OF PHYSICAL AND COMPUTATIONAL SCIENCE



**PROJECT DOCUMENTATION
FOR CONSIDERATION UNDER
BSc. COMPUTER SCIENCE**

**PROJECT TITLE
AIDBRIDGE: DISASTER RELIEF DONATION PLATFORM**

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DECLARATION

We hereby declare that this Project submission is our work toward the award of the BSc. Computer Science, and that we have wholly undertaken the AidBridge project as a final year project and reported under the supervision of Dr. Benjamin Parthey and that to the best of knowledge it contains no material previously published by another person nor material which has been accepted for the award of any degree of the university, except where the due acknowledge and references has been made in the document.

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RENEE deGRAFT-ETSISON

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DATE

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MICHAEL OBENG

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DATE

I declare that I have supervised Renee deGraft-Etsison and Michael Obeng in undertaking this Project (AidBridge web application) and confirm its submission for assessment.

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DR. BENJAMIN PARTEY

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DATE

ABSTRACT

AidBridge is an innovative disaster relief donation platform designed to streamline the connection between donors and individuals or communities impacted by disasters. Recognizing the critical need for rapid, transparent, and accountable humanitarian support, the platform integrates modern web technologies, including HTML, CSS, JavaScript, Firebase, and Google Vision API, to deliver a secure and user-friendly experience. AidBridge not only simplifies the donation process but also introduces a verifiable request system that ensures that aid reaches the most affected victims based on real needs.

Through features such as real-time request tracking, categorized donation requests, and multimedia questionnaires, the platform provides both donors and administrators with greater confidence in the integrity of the aid process. The use of Firebase enables scalable backend data management, while EmailJS and Twilio API facilitates seamless communication between parties. The inclusion of the Google Vision API offers an additional verification layer by analyzing uploaded images for authenticity and relevance.

Developed using agile methodologies, AidBridge was built through iterative sprints involving continuous evaluation, user feedback, and system enhancements. This documentation presents a comprehensive overview of the system development life cycle, including requirement analysis, design specifications, system implementation, testing strategies, and evaluation outcomes, providing valuable insight into the creation and deployment of a responsive and effective digital disaster response solution.

DEDICATION

We dedicate this project, "AidBridge Web Application," to our families, computer science department and friends, whose unwavering support and encouragement have been our greatest source of strength throughout this journey.

To our parents, who have always believed in us and provided the foundation for our academic and personal growth. Your love, sacrifices, and endless support have been the driving force behind our success.

To our project supervisor, lecturers and the staff at the computer science department, who has nurtured us, encouraged us and created an optimistic learning environment to throughout this journey. Your support has been key to our success.

To our friends, who have stood by us through thick and thin, offering constant motivation and understanding. Your solidarity and encouragement have been invaluable.

Finally, we dedicate this project to each other. Our partnership has been a testament to teamwork, perseverance, and mutual respect. Together, we have overcome challenges, celebrated successes, and grown both personally and professionally.

Thank you for being a part of this remarkable journey with us.

ACKNOWLEDGEMENTS

We would like to express our heartfelt gratitude to all those who have supported and guided us throughout the development of my project, "AidBridge Web Application."

Firstly, we extend our sincere appreciation and thanks to our project supervisor, Dr. Benjamin Partey, for his invaluable guidance, encouragement, and constructive feedback on the project. Your insights and expertise have been helpful in shaping this project and have greatly enhanced our learning and practical experience.

We are also deeply grateful to Kwame Nkrumah University of Science and Technology for providing the resources and environment necessary for this project. The access to advanced facilities and a supportive academic community has been crucial to our success.

Special thanks to the computer science department, the lecturers, the staff, our peers and colleagues, whose collaboration and moral support have been a great source of motivation and inspiration. We are also thankful to our families and friends for their unwavering support and understanding during the course of this project.

Finally, we would like to acknowledge the open-source community and various online resources that provided the resources, tools and frameworks used in this web application. Your contributions have been invaluable in bringing this project to fruition.

This project is a reflection of the collective effort and support from many individuals, the department and organizations, and we are deeply appreciative of everyone who has played a part in this journey.

CHAPTER I

INTRODUCTION

1.1 Background of the Study.

Disasters, whether natural or man-made, have increasingly disrupted the lives of millions of people globally, leaving communities vulnerable and in urgent need of assistance. These events often create situations where victims require immediate support in the form of food, funds, medical supplies, and shelter. However, challenges such as poor coordination, lack of transparency, and the difficulty of verifying genuine beneficiaries hinder the effectiveness of traditional disaster relief efforts.

In Ghana and other developing countries, these challenges are even more pronounced, as limited resources and logistical constraints make it difficult to ensure aid reaches those most in need. At the same time, many individuals and organizations are willing to donate, but the absence of a structured and transparent platform prevents these donations from being effectively distributed.

AidBridge: Disaster Relief Donation Platform was conceived to address this problem by bridging the gap between donors and disaster victims. The platform leverages modern web technologies to create a centralized, secure, and transparent system for disaster relief. Victims can request resources such as food, medical kits, or clothing, while donors can easily identify needs and respond appropriately. To promote accountability, the system introduces an intelligent questionnaire powered by an algorithm that evaluates user responses to assess proof-of-need and provisions for future integration with the Ghana Card verification system via API access to the National Identification Authority (NIA).

By combining efficient technology, real-time data management, and verification mechanisms, AidBridge seeks to enhance the credibility, efficiency, and fairness of disaster relief distribution, ultimately improving the resilience and recovery of affected communities.

1.2 Problem Statement

The increasing frequency and severity of natural and man-made disasters have left millions of people vulnerable, often without access to essential resources such as food, shelter, clean water, and medical supplies. Despite the presence of local and international donors willing to provide aid, the distribution process is frequently plagued by challenges including lack of transparency, logistical inefficiencies, and difficulties in verifying the eligibility of beneficiaries. These issues result in delays, unequal distribution, and in some cases, the diversion of aid away from those most in need.

The problem is further compounded by the absence of a centralized, user-friendly platform that connects donors directly with victims of disasters. Currently, many relief efforts rely on fragmented systems or manual processes that limit accountability and hinder real-time communication between stakeholders. This creates uncertainty for donors who wish to ensure their contributions are making a real impact, while victims often struggle to prove their needs or access the resources being distributed.

The social impact of this problem is significant: disaster victims are left without adequate and timely support, which can worsen their vulnerability and slow their recovery. Economically, inefficiencies in the aid distribution process result in wasted resources and donor fatigue, as well-meaning contributors lose confidence in existing systems.

Therefore, there is a critical need for a centralized disaster relief platform that ensures efficient aid distribution, enables transparent verification of victims, and fosters trust between donors and beneficiaries. AidBridge seeks to address this gap by providing a digital platform that streamlines the request, donation, and verification processes, ultimately ensuring that disaster relief reaches the truly needy in a timely and accountable manner.

1.3 Aim of the Project.

The primary aim of the AidBridge platform is to create an efficient, transparent, and userfriendly disaster relief donation system that connects donors with victims and communities in need. By doing so, we aim to improve the efficiency of aid distribution, foster trust and accountability in the donation process, and ensure that resources reach the truly needy on time.

1.4 Objectives of the Project.

The specific objectives of the AidBridge project are to:

1. *Facilitate Disaster Relief Donations* by providing an intuitive interface for individuals and organizations to donate essential items such as food, clothing, medical supplies, and shelter materials.
2. *Enhance Beneficiary Access to Aid* by enabling victims to request specific items needed during disasters through a simple and accessible platform.
3. *Promote Verification and Accountability* by integrating Ghana Card identification, profile pictures, questionnaires, and exploring the future use of APIs (e.g., Ghana NIA) to verify the authenticity and needs of applicants.
4. *Streamline the Distribution* of donated items by offering a centralized system that reduces duplication, eliminates inefficiencies, and ensures fair allocation of resources.
5. *Encourage Donor Confidence and Engagement* by providing transparency in donation processes, including proof of receipt, real-time tracking, and feedback mechanisms.
6. *Support Long-Term Resilience* by offering a sustainable system that can be adapted for future disaster relief efforts beyond immediate emergencies.

1.5 Scope of the Project

The scope of this project defines the boundaries of the AidBridge Disaster Relief Donation Platform and highlights the core areas it covers. The platform is designed to serve as a bridge between donors and victims of disasters by focusing on donation management, beneficiary request processing, verification, and administrative oversight. It incorporates a secure registration and authentication system for both donors and beneficiaries, ensuring that only authorized users can access and interact with the platform. Donors can contribute essential relief items, such as food, clothing, medical kits, and personal care products, while providing key details, including type, quantity, and description, for effective categorization.

Beneficiaries, on the other hand, can submit specific requests for items they need, including details such as food preferences or medical requirements. To ensure accountability and fairness, the system requires beneficiaries to upload a valid Ghana Card, a profile picture, and to complete a questionnaire that captures their occupation, age, income level, and health status. This verification mechanism helps categorize and prioritize applicants based on

their level of need. Future integration with the Ghana NIA API has also been envisioned to allow automatic verification of Ghana Card details.

The platform further includes a notification system powered by EmailJS and Twilio API, which sends automated updates to users regarding donations, requests, approvals, and distribution activities. Finally, an admin dashboard is provided to enable system administrators to review beneficiary applications, verify uploaded documents, manage user accounts, and generate reports for transparency and accountability.

1.6 Limitations of the Project

Despite its broad coverage, the AidBridge platform has several limitations.

First, while the system manages requests and donations digitally, it does not handle the physical delivery or logistics of donated items. Beneficiaries are expected to claim their approved items from designated distribution points, which may limit accessibility for those in remote areas.

Second, although the project envisions integration with the Ghana NIA API to automatically verify Ghana Card details, this feature could not be implemented due to time and resource constraints. As a result, verification relies on manual review of uploaded documents and questionnaire responses.

Finally, as this is a prototype system, scalability and real-world stress testing were limited, meaning performance and reliability under large-scale disaster scenarios remain areas for future improvement.

1.7 Approach to the Project

The development of the AidBridge Disaster Relief Donation Platform followed a systematic and structured approach, ensuring that the system met both donor and beneficiary needs while maintaining transparency and accountability. The key stages were as follows:

1. *Requirement Analysis*

Research was conducted to understand the challenges faced by disaster victims, donors, and administrators. This included reviewing existing relief systems, studying similar platforms, and identifying gaps that AidBridge could fill. Key requirements such as donation management, beneficiary verification, and transparency mechanisms were prioritized.

2. *Design and Planning*

The design process involved creating wireframes for user interfaces, database schemas for storing donations and requests, and diagrams such as use case and activity diagrams to illustrate system workflows. A project timeline with milestones was also established, guided by Agile methodologies.

3. *Development*

The platform was developed using HTML, CSS, and JavaScript for the frontend, with Firebase for authentication and data storage. EmailJS and Twilio API were integrated for notifications, while the Google Vision API was explored for proof-of-need verification. The system was developed iteratively, with core features like user registration and donation requests implemented first, followed by additional enhancements.

4. *Testing*

Extensive testing was conducted at each stage, including unit testing of individual components, integration testing of connected modules, and user acceptance testing with sample users. This ensured the application was secure, functional, and userfriendly.

5. *Deployment*

The system was deployed on a scalable cloud platform, allowing accessibility from different devices. Deployment included configuration for authentication and secure storage of uploaded documents.

6. *Evaluation and Feedback*

Initial evaluations with test users provided valuable feedback on usability, clarity of processes, and areas needing improvement. Adjustments were made to the request verification process and notification features to enhance effectiveness.

7. *Documentation*

Comprehensive documentation was prepared to detail the project's requirements, design, development, testing, and results. This documentation serves as a record for developers, supervisors, and stakeholders, while also acting as a reference for future improvements.

1.8 Development Timeline

The AidBridge Disaster Relief Donation Platform was developed following a phased approach, with specific tasks allocated to each stage of the project.

The timeline below summarizes the major phases and their respective activities:

Phase	Tasks	Duration
1. Planning	- Conducted initial research to define system requirements. - Gathered requirements through literature review, online surveys, and stakeholder discussions. - Assessed feasibility of the proposed project.	Week 1 - 4
2. Design	- Created detailed project plan and milestones. - Designed system architecture using diagrams (use-case, activity, and flowcharts). - Developed wireframes and prototypes for the user interface.	Week 5 - 6
3. Development	- Implemented the frontend using HTML, CSS, and JavaScript. - Developed backend with Firebase for authentication and data storage. - Initiated EmailJS, Twilio API, and Google Vision API integration. - Set up and linked the system database.	Week 7 - 21
4. Testing	- Conducted unit, integration, and system testing. - Validated request verification, notifications, and data storage. - Ensured system security and optimized performance.	Week 22 - 24
5. Evaluation	- Deployed system on a cloud platform. - Conducted initial evaluation with test users to collect feedback. - Adjusted features based on feedback.	Week 25 - 27
6. Documentation	- Compiled and refined project documentation. - Prepared technical details and final project report.	Week 28 - 29
7. Finalization	- Prepared project presentation. - Conducted demonstration of the system features.	Week 30

CHAPTER II

REVIEW OF LITERATURE AND TOOLS

2.1 Background Review

Vulnerability and access to aid remain significant global challenges, affecting millions of individuals and communities worldwide. Despite advances in disaster management, healthcare, and humanitarian support, issues such as poverty, unemployment, health crises, and disasters continue to hinder access to essential resources. In Ghana and across the globe, many individuals face daily struggles to secure food, clothing, medical supplies, and other basic needs.

At the same time, inefficiencies in aid distribution and verification often result in delays, duplication of efforts, and in some cases, resources not reaching the most vulnerable. This highlights the urgent need for innovative solutions that ensure transparency, accountability, and accurate targeting of aid beneficiaries.

The AidBridge platform was developed to bridge this gap. It provides a structured way for donors to contribute essential items (food, clothing, medical kits, personal care products, and tools) and for beneficiaries to request the support they need. AidBridge incorporates a questionnaire-based categorization system (covering occupation, income, age, and health) to help identify the most vulnerable and ensure resources are distributed fairly.

In line with the **United Nations Sustainable Development Goals (SDGs)**, especially SDG 1 (*No Poverty*) and SDG 2 (*Zero Hunger*), AidBridge demonstrates how web-based platforms can play a vital role in addressing poverty, inequality, and disaster relief by connecting donors directly with verified beneficiaries.

2.2 Review of Existing Systems

Several platforms and applications have been developed to facilitate humanitarian aid, donations, or community support. Studying these systems provides insight into their strengths, weaknesses, and lessons for AidBridge.

- *Food Rescue US*: A U.S.-based system that rescues surplus food and delivers it to social service agencies. While user-friendly, its reliance on volunteers and limited geographic reach constrain its scalability.
- *Olio*: A mobile app that enables neighbors and businesses to share surplus food. It promotes community engagement but requires high user participation and manual coordination of exchanges.
- *Too Good To Go*: An app connecting users with discounted surplus food from restaurants and stores. It effectively reduces food waste but is commercially focused and less accessible to the most vulnerable.
- *Feeding America & MealConnect*: Large-scale platforms that coordinate food banks and donors. They operate effectively at scale but rely heavily on extensive networks and resources, which can limit adaptation to smaller communities or broader aid beyond food.

2.3 Gap Identified: Existing systems mostly focus on **food donation** or **waste reduction**, but very few integrate comprehensive aid (food, clothing, health supplies, etc.) with verification systems to ensure fairness and accountability. AidBridge addresses this gap by offering multi-category aid support and an algorithm-driven questionnaire system verification of need.

2.4 Problem Identification of Existing Systems

From the review above, key problems persist:

- *Dependency on volunteers* leads to inconsistent distribution.
- *Geographic limitations* prevent global or national reach.
- *Commercial focus* excludes the most vulnerable populations.
- *Lack of verification* systems risks unfair allocation of resources.
- *Narrow scope (mostly food-only)* fails to meet the holistic needs of victims and vulnerable populations.

AidBridge overcomes these challenges by:

- Expanding donations beyond food to include clothing, medical kits, personal care, and tools.
- Integrating a questionnaire to assess vulnerability (income, occupation, health, age).
- Incorporating Ghana Card upload and potential future NIA API verification.
- Using web technologies (Firebase, EmailJS, Twilio API, Google Vision API) to ensure secure, transparent operations.

2.5 Project Evaluation

Compared to the existing systems:

- *Integration and Coordination:* AidBridge integrates donors, beneficiaries, and administrators with built-in verification, making coordination smoother.
- *Scalability:* Designed as a web app, it can be scaled across regions and communities without heavy reliance on physical networks.
- *User Verification:* Unlike existing systems, AidBridge uses questionnaires and ID uploads for fairness in resource distribution.
- *Multi-Sector Aid:* Goes beyond food to include health, clothing, tools, and personal care items.
- *Transparency & Trust:* Admin dashboard ensures accountability through monitoring and reporting.

2.6 Review of Project-Related Methodologies

AidBridge's design draws from best practices of the reviewed systems:

- *Agile Development:* Iterative sprints with regular feedback.
- *User-Centred Design:* Simple UI for donors and beneficiaries.
- *Cloud-Based Architecture:* Firebase backend ensures scalability, real-time updates, and security.
- *Data Security and Privacy:* Ghana Card, questionnaire data, and personal info are stored securely.
- *Community Engagement:* Encourages local and international donors to directly impact lives.
- *Continuous Evaluation:* Feedback from users and administrators ensures ongoing improvement.

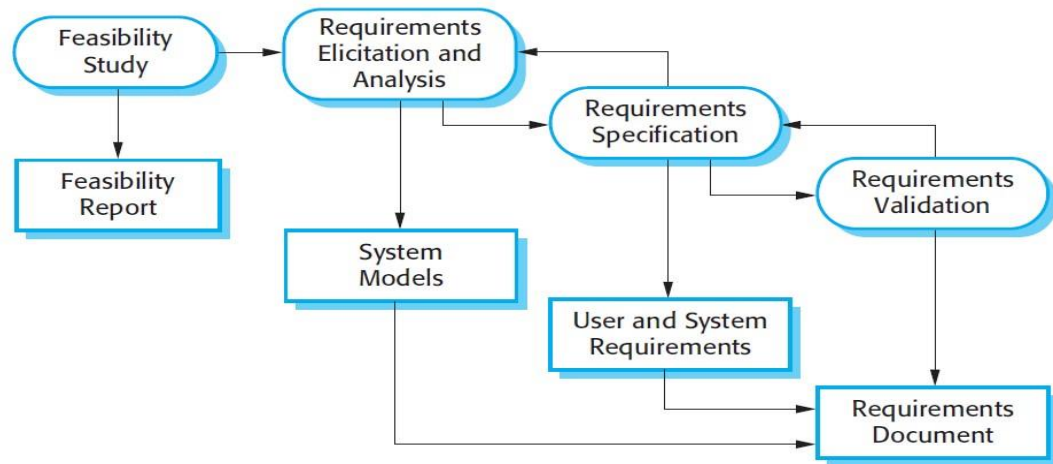
CHAPTER III

REQUIREMENTS SPECIFICATIONS

3.1 Requirement Gathering

AidBridge's requirements were gathered through:

- *Stakeholder Interviews*: Discussions with potential donors, NGOs, community leaders, and beneficiaries.
- *Surveys & Questionnaires*: Distributed to gather data on what people most need in times of disaster or vulnerability.
- *Focus Groups*: Sessions with disaster victims, low-income families, and local relief workers.
- *Observation*: Examined existing donation systems and distribution bottlenecks.
- *Document Review*: Studied related platforms, UN SDGs, and academic works on aid systems.



3.2 Functional Requirements of AidBridge

1. Donation of Relief Items

- Donors can donate essential relief items (food, clothing, medical kits, personal care products, etc.) by finding our locations on the platform and visiting in person to donate.

- Each donation entry must include item details such as type, description, and quantity.
2. *Request for Aid*
 - Beneficiaries can request specific items (e.g., food type, clothing, medical kits).
 - Requests will be linked to beneficiary verification (algorithm-driven questionnaire + Ghana Card).
 3. *Search and Access Donated Items*
 - Beneficiaries can search for available donations by category, item type, or location.
 - System displays matching results and availability status.
 4. *User Verification (Proof of Need)*
 - Beneficiaries complete a questionnaire (age, occupation, income, health, etc.) which the system process with an algorithm to assess their eligibility.
 - Uploading of Ghana Card and profile picture for identity verification.
 - Future integration with Ghana NIA API for automatic validation.
 5. *Donor Support for Platform Growth*
 - Donors can contribute funds to AidBridge to support operational costs and improve system efficiency.
 6. *Contact and Communication*
 - Users can contact AidBridge administrators via email or integrated contact form.
 - Automated notifications (Email/SMS) confirm successful donations, requests, and approvals.
 7. *Access to Educational Resources*
 - Users can access blogs, articles, and resources on disaster preparedness, relief management, and community support.
 8. *Donation and Request Confirmation*

- Donors receive confirmation when their fundings are successfully received.
- Beneficiaries receive confirmation when their requests are approved and allocated.

9. *Admin Monitoring*

- Administrators can review requests, verify documents, approve/reject applications, and generate reports.

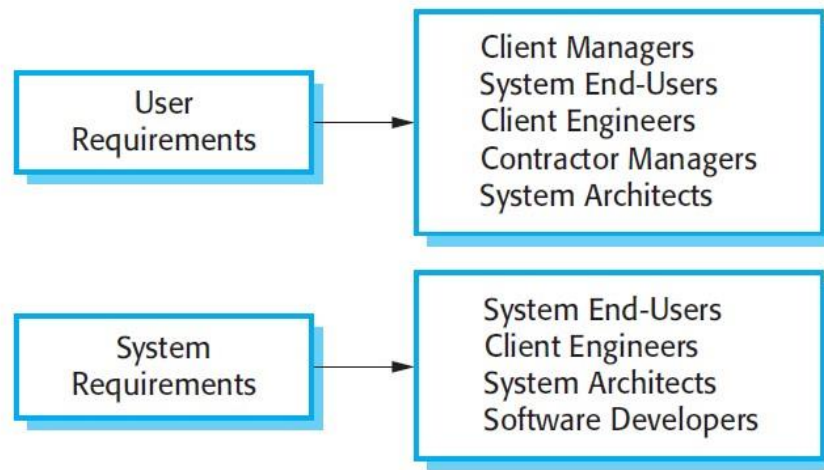
3.3 Functional User Requirements (AidBridge)

1. User should be able to use the platform to find AidBridge warehouse locations on a map, where they can physically donate essential relief items (food, clothing, medical kits, personal care items, etc.).
2. User should be able to request specific relief items (e.g., clothing, food type, medical supplies) from the system.
3. The user should be able to search for available donations in the platform's database or linked relief centres.
4. The user should be able to verify their identity by uploading a Ghana Card, profile picture, and completing an algorithm-driven questionnaire (occupation, age, health, income, etc.).
5. The user should be able to contact AidBridge administrators through email or an integrated messaging system.
6. The user should be able to send funds/donations to AidBridge to support platform improvement and operational costs.
7. The user should be able to access educational resources and blogs on disaster preparedness, aid distribution, and relief efforts.
8. The user should be able to receive notifications and confirmations (via Email/SMS) for successful donations, requests, approvals, or rejections.

3.4 Functional System Requirements (AidBridge)

1. User shall enter the domain of the website.
2. System shall display the registration page.
3. User shall enter their full name, email, password, and click the Sign Up button.
4. System shall verify and validate the user's information.
5. System shall store user information in the database.
6. System shall display the login page.
7. User shall enter their email and password and click the Login button.
8. System shall validate the entered credentials.
9. System shall display the home page.
10. User shall select the Fund Us button to contribute funds.
11. System shall display a payment form.
12. User shall fill in the input fields (email address, amount, first name, and last name) and submit.
13. System shall verify and validate the funding details.
14. System shall store the donated item details in the database.
15. System shall send a confirmation notification (Email) to the donor.
16. User shall select the Request button to ask for aid.
17. System shall display a request form linked to the questionnaire and Ghana Card upload.
18. User shall fill in the request details and verification info and submit.
19. System shall verify and validate the request and supporting documents.
20. System shall store the request information in the database.
21. Admin shall review the beneficiary's request and approve or reject it.
22. System shall send a notification (Email/SMS) to the user about the status of their request.
23. If approved, system shall provide instructions on how and where the aid will be distributed.

24. System shall allow beneficiaries to search donations by category (food, clothing, medical kits, etc.) and view availability.
25. If an item is unavailable, the system shall display a “not available” message.
26. System shall allow admin to generate reports on donations, requests, and approvals for transparency.
27. User shall be able to log out securely from the platform.



3.5 Non-Functional Requirements

- *Performance:*

The system must be capable of handling a high volume of concurrent users (donors, beneficiaries, and administrators) and manage their operations without significant degradation in performance. Response times for user actions such as login, donation submission, or request approval should be minimal.

- *Scalability:*

The system must be scalable to accommodate growth in the number of users, donations, requests, and expansion into multiple geographic regions. This includes the ability to handle increased data storage and processing needs.

- *Security:*
The system must implement robust security measures to protect user data, including Ghana Card details, questionnaire results, and donation records. This includes encryption, secure authentication mechanisms, and regular security audits.
- *Reliability:*
The system must remain reliable and highly available with minimal downtime.
Failover mechanisms, backup procedures, and disaster recovery protocols should be included to ensure continuity of service.
- *Usability:*
The system must be user-friendly and intuitive, with a clear and consistent interface design that works well for both donors and beneficiaries. It should also be accessible to users with varying levels of technical literacy.
- *Compatibility:*
The platform must be compatible with major web browsers (Chrome, Firefox, Safari, Edge) and adaptable to desktops, tablets, and smartphones.
- *Maintainability:*
The system must be designed for easy maintenance and updates.
This includes modular code development, comprehensive technical documentation, and automated testing pipelines.
- *Compliance:*
The system must comply with relevant legal and regulatory requirements, including data protection and privacy laws (e.g., Ghana Data Protection Act, GDPR principles).

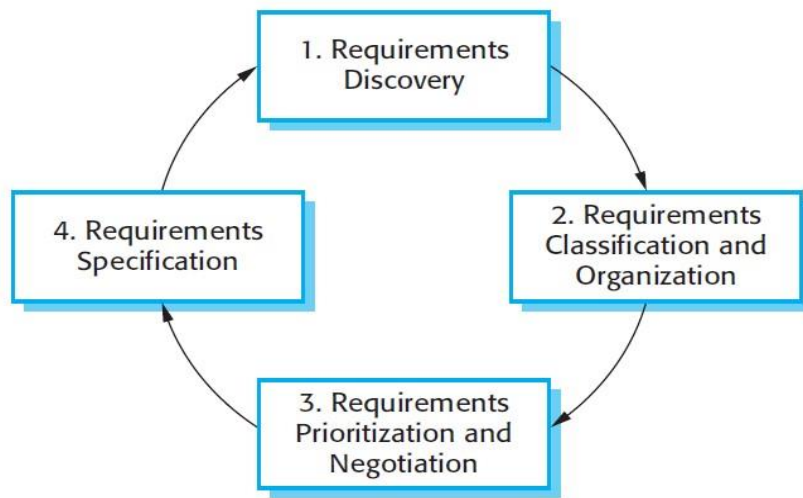
3.6 Hardware Requirements (AidBridge)

To ensure efficient hosting and operation of the AidBridge platform, the following infrastructure is required:

- *Web Server:*
A robust web server capable of handling high traffic and concurrent user requests. Recommended specifications include:
 - Processor: Multi-core (e.g., Intel Xeon or AMD EPYC)
 - RAM: Minimum 16 GB (scalable based on traffic)
 - Storage: SSD for fast read/write operations
 - Network: High-speed internet connection with redundancy
- *Database Server:*
A dedicated server for storing and managing user data, donation records, requests, and questionnaire responses. Recommended specifications include:
 - Processor: Multi-core
 - RAM: 32 GB (to handle concurrent transactions)
 - Storage: SSD with backup capability
 - Network: High-speed, secure connection
- *Networking Equipment:*
Reliable networking hardware (routers, switches, and firewalls) to ensure secure and efficient data transmission.
- *Load Balancer:*
To distribute incoming traffic across multiple servers, ensuring optimal performance and preventing overloads.
- *Security Hardware:*
Hardware Security Modules (HSM) for secure key management, encryption, and compliance with data protection standards.

3.7 Requirements Analysis (AidBridge)

- *Feasibility Study:*
Assessing the technical, operational, and economic feasibility of AidBridge requirements. This includes evaluating the availability of technologies (Firebase, Google Vision API, EmailJS, Twilio API), development skills, and infrastructure to implement the system effectively.
- *Prioritization:*
Requirements are prioritized based on impact and urgency. For example, user registration, donation management, and beneficiary verification are high-priority, while advanced analytics and API integration with the Ghana NIA are secondary features for future development.
- *Validation:*
Requirements are validated with stakeholders (donors, beneficiaries, supervisors, and system administrators) to ensure they accurately reflect real-world needs and expectations. Reviews, prototypes, and testing sessions are used for validation.
- *Traceability:*
Each requirement is linked to specific project deliverables and test cases to ensure complete implementation during development and testing.
- *Conflict Resolution:*
Potential conflicts between donor needs, beneficiary expectations, and technical constraints are identified and resolved early. Trade-offs are negotiated and agreed upon with stakeholders to maintain balance and feasibility.



3.8 UML USE CASE DIAGRAM OF THE PROJECT

Actors

- *Donor*: A user who contributes essential relief items (e.g., food, clothing, medical kits, personal care products).
- *Beneficiary (Recipient)*: A user who requests and receives aid through the platform by providing personal details, Ghana Card verification, and completing the eligibility questionnaire.
- *Administrator*: A user who manages the platform, verifies beneficiary applications, oversees donation and request processes, and generates reports for transparency.

Key Use Cases

1. *Donor*

- Visit the donation platform
- Donate any amount they can afford
- Receive Confirmation Notifications

OR

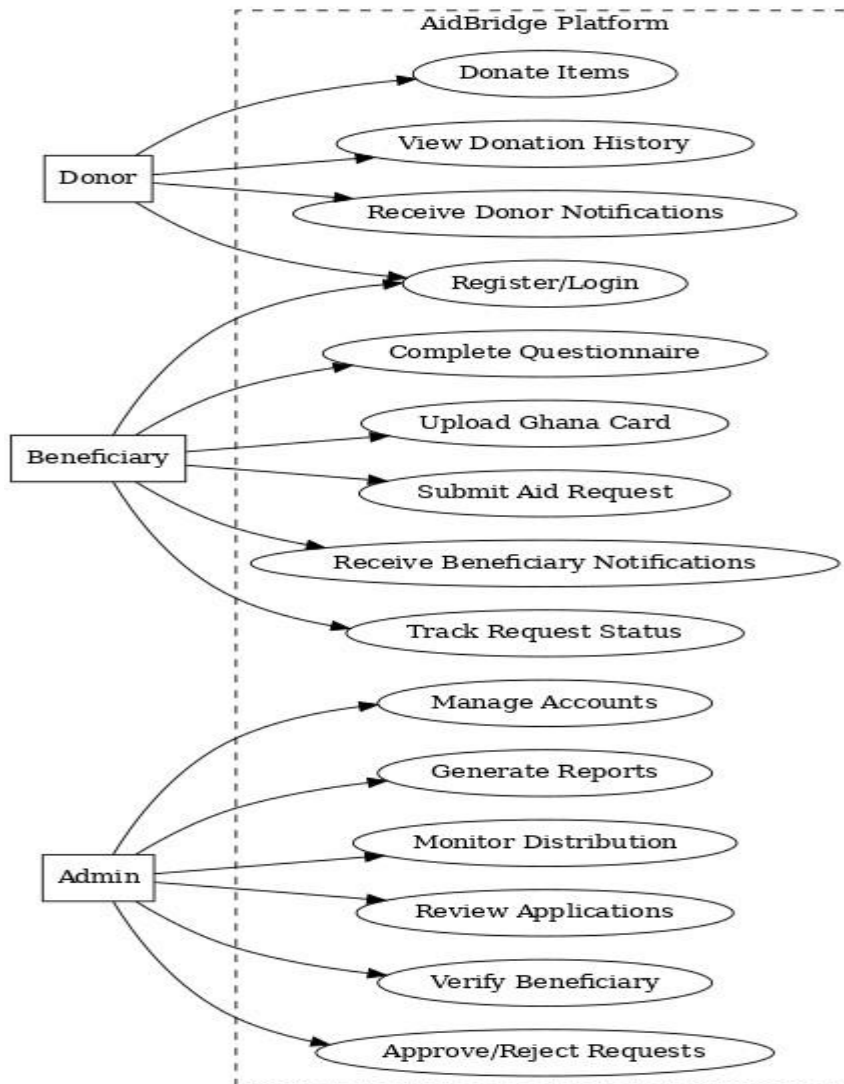
- Visit the donation platform
- Go to donate to find our locations closer to them
- Deliver items they want to donate to the location

2. *Beneficiary (Recipient)*

- Register / Log in
- Complete an algorithm-driven questionnaire (occupation, age, health, income)
- Upload Ghana Card & Profile Picture
- Submit Aid Request
- Receive Approval/Denial Notifications
- Track Request Status

3. *Administrator*

- Review Beneficiary Applications
- Verify Ghana Card & Questionnaire Responses
- Approve/Reject Aid Requests
- Manage Donors and Beneficiaries
- Generate System Reports
- Monitor Donations & Distribution



CHAPTER IV

DESIGN SPECIFICATIONS

4.1 System Design and Methodology

The *Agile Methodology* was adopted for the development of the AidBridge Disaster Relief Donation Platform to allow for flexibility, adaptability, and continuous feedback during the system's lifecycle. Agile enabled the team to work in short iterations (sprints), incorporate user and supervisor feedback, and progressively enhance the system with each cycle.

4.2 Architectural Design

The AidBridge platform is designed using a multi-tiered architecture, ensuring modularity, scalability, and maintainability. Each layer has a distinct role in managing the system's functionality:

1. *Presentation Layer*

- Manages user interaction through a responsive, intuitive web interface.
- Developed with HTML, CSS, JavaScript, and frameworks for enhanced UI/UX.
- Includes forms for registration, login, donation requests, questionnaire submissions, and proof-of-need uploads.

2. *Application Layer*

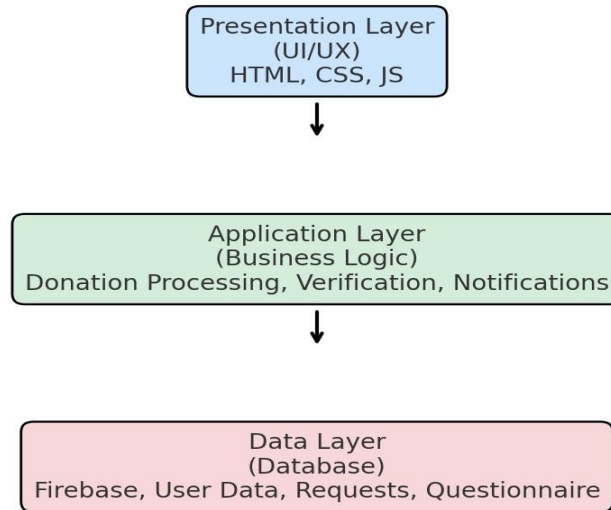
- Contains the business logic of the platform, handling workflows such as donation processing, request verification, and notification triggers.
- Facilitates communication between the presentation layer and the data layer.
- Implements rules for categorizing users based on questionnaire responses (age, health, income, occupation).
- Includes the integration of services like Firebase Authentication, EmailJS and Twilio API for secure communication and alerts.

3. *Data Layer*

- Responsible for *secure storage and management* of all system data.

- Stores user accounts, donation records, requests, questionnaire results, and verification documents.
- Built using Firebase Realtime Database/Firestore for scalability, real-time updates, and cloud-based accessibility.
- Ensures data privacy and security through encryption and authentication protocols.

AidBridge System Architecture (3-Tier Model)



4.3 Technology Stack

The technology stack chosen for the *AidBridge Disaster Relief Donation Platform* includes:

FRONT-END

- *HTML5*: For creating the structure of the web application.
- *CSS3 / Bootstrap 5*: For styling web pages and ensuring responsiveness across devices.
- *JavaScript*: For adding interactivity and client-side scripting.

BACK-END

- *Firebase*: Provides backend services for authentication, hosting, and real-time data storage.
- *EmailJS and Twilio API*: Used to send automated email notifications to users.

- *Google Cloud Vision API (optional/future extension)*: For document and image verification.
- *Node.js + Express.js (optional for scalability)*: To handle API requests and backend logic where needed.

DATABASE

- *Firebase Realtime Database*: A NoSQL cloud database for storing user information, donations, requests, and verification data.

4.4 Other Tools and APIs

- *Google APIs*: API for building web maps to show donation centres, and beneficiary locations.
- *SweetAlert2*: Provides customizable, responsive popup boxes for user notifications.
- *Ghana NIA API (Future Integration)*: Planned integration to verify user identities through Ghana Card validation.
- *Git/GitHub*: For version control and collaborative development.

4.5 Module Decomposition

The AidBridge platform is divided into several modules, each responsible for specific functionalities:

1. *User Management Module*
 - Handles user registration and authentication via Firebase.
 - Manages donor, beneficiary, and administrator roles.
2. *Funding Management Module*
 - Allows donors to contribute any amount they can to the platform
 - Records donation details.
3. *Beneficiary Request & Verification Module*
 - Enables victims/beneficiaries to request needed items.
 - Uses uploaded Ghana Card, profile picture, and questionnaire (occupation, age, income, health, etc.) to categorize and verify beneficiaries.
4. *Administrator Management Module*
 - Allows administrators to review requests and donations.
 - Manages user verification, approves/rejects beneficiaries, and ensures transparent resource distribution.
5. *Notification Module*
 - Sends real-time notifications via EmailJS (emails) and Twilio API (SMS) integration.

- Keeps donors and beneficiaries updated on approvals, allocations, and deliveries.
6. *Mapping & Location Module*
- Uses Google Maps API to display donation centres, collection points, and affected areas.
 - Helps donors and beneficiaries identify nearby centres.
7. *Reporting and Transparency Module*
- Provides reports on donation volumes, verified beneficiaries, and aid distribution.
 - Enhances accountability for donors and administrators.

4.6 System Interfaces

User Interface (UI)

- Web pages designed with HTML5, CSS3, Bootstrap, and JavaScript.
- Forms for user input (registration, login, donation, requests, questionnaire, etc.).

Application Programming Interface (API)

- Firebase Authentication API for secure user management.
- EmailJS and Twilio API for email and SMS notifications.
- Ghana NIA API (planned) for future ID verification.

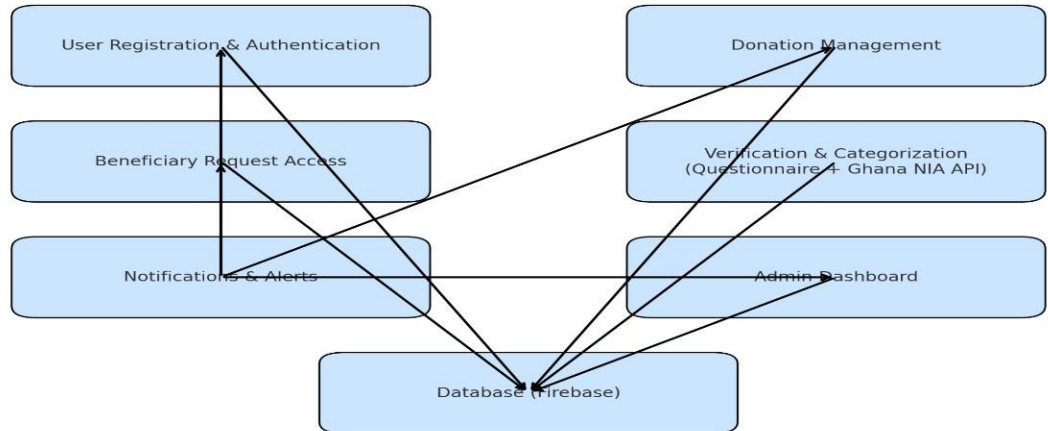
Database Interface

- Firebase Realtime Database for real-time read/write operations.

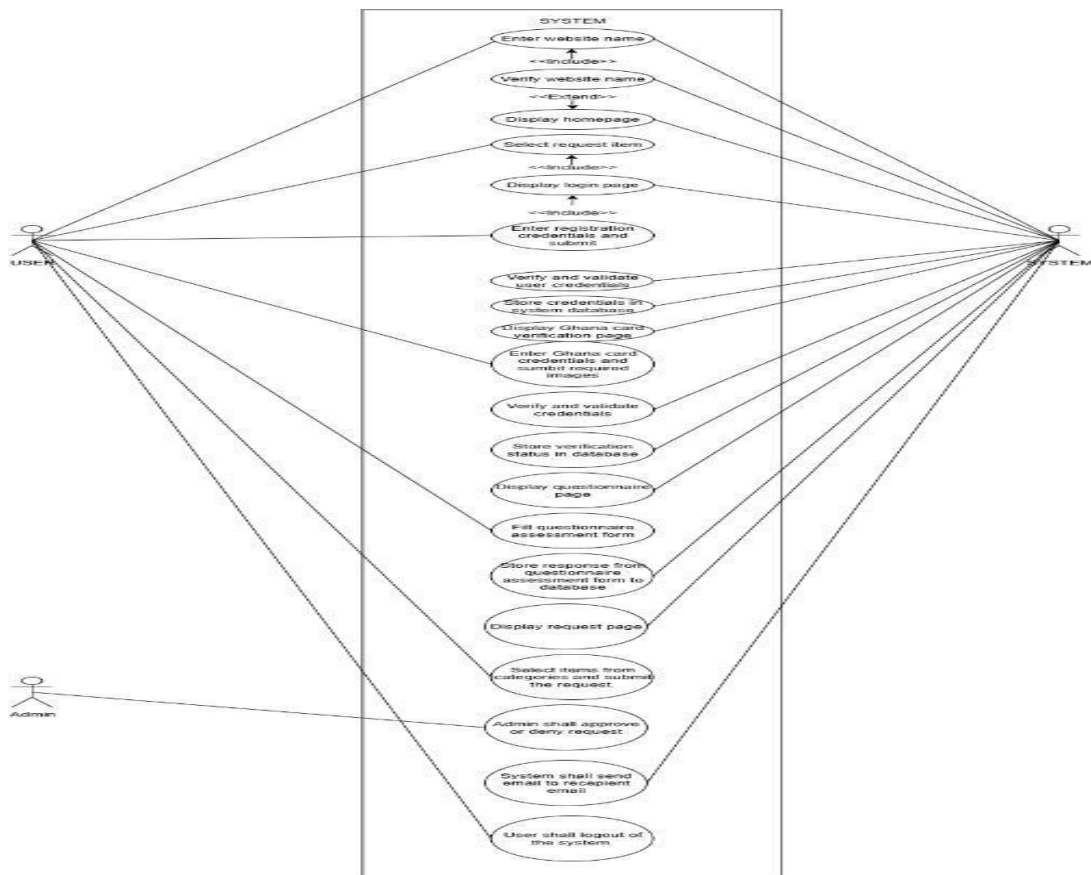
External Services Interface

- Google apis for maps.
- SweetAlert2 for enhanced UI alerts.
- Email and SMS gateway (e.g., EmailJS, Twilio API) for notifications.

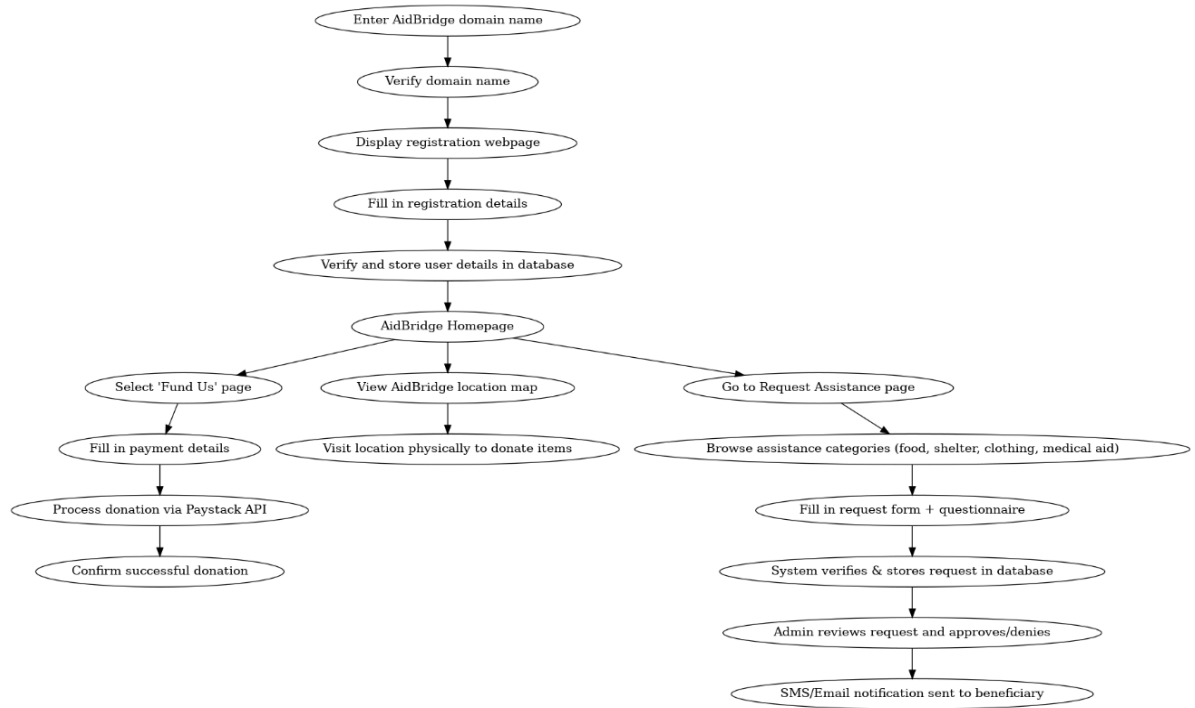
AidBridge System Module Interaction Diagram



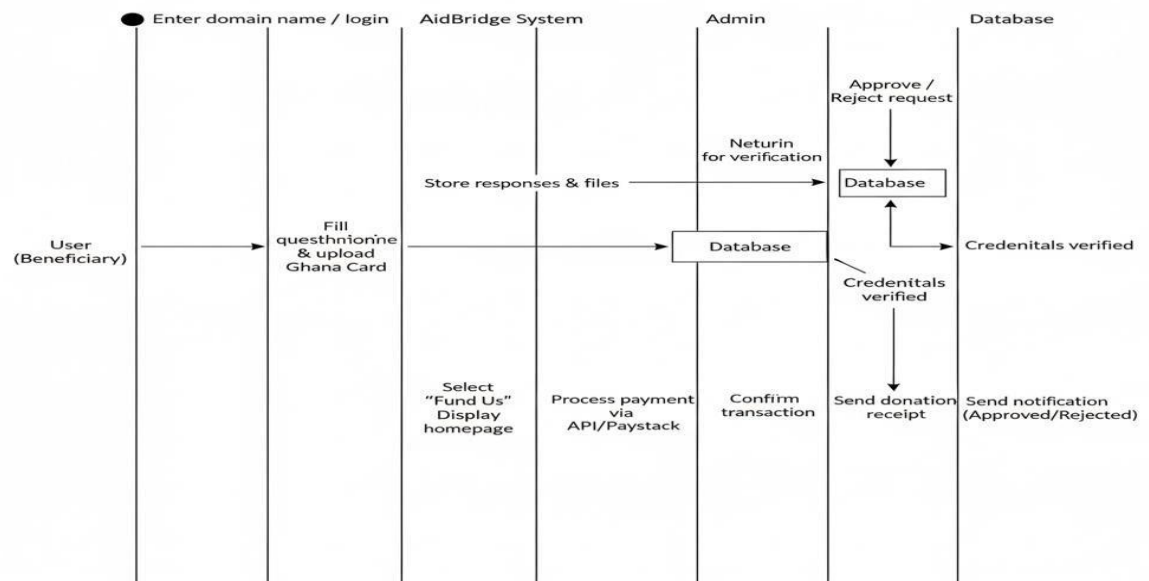
4.7 Requirement Model (Uml Use Case Diagram Description)



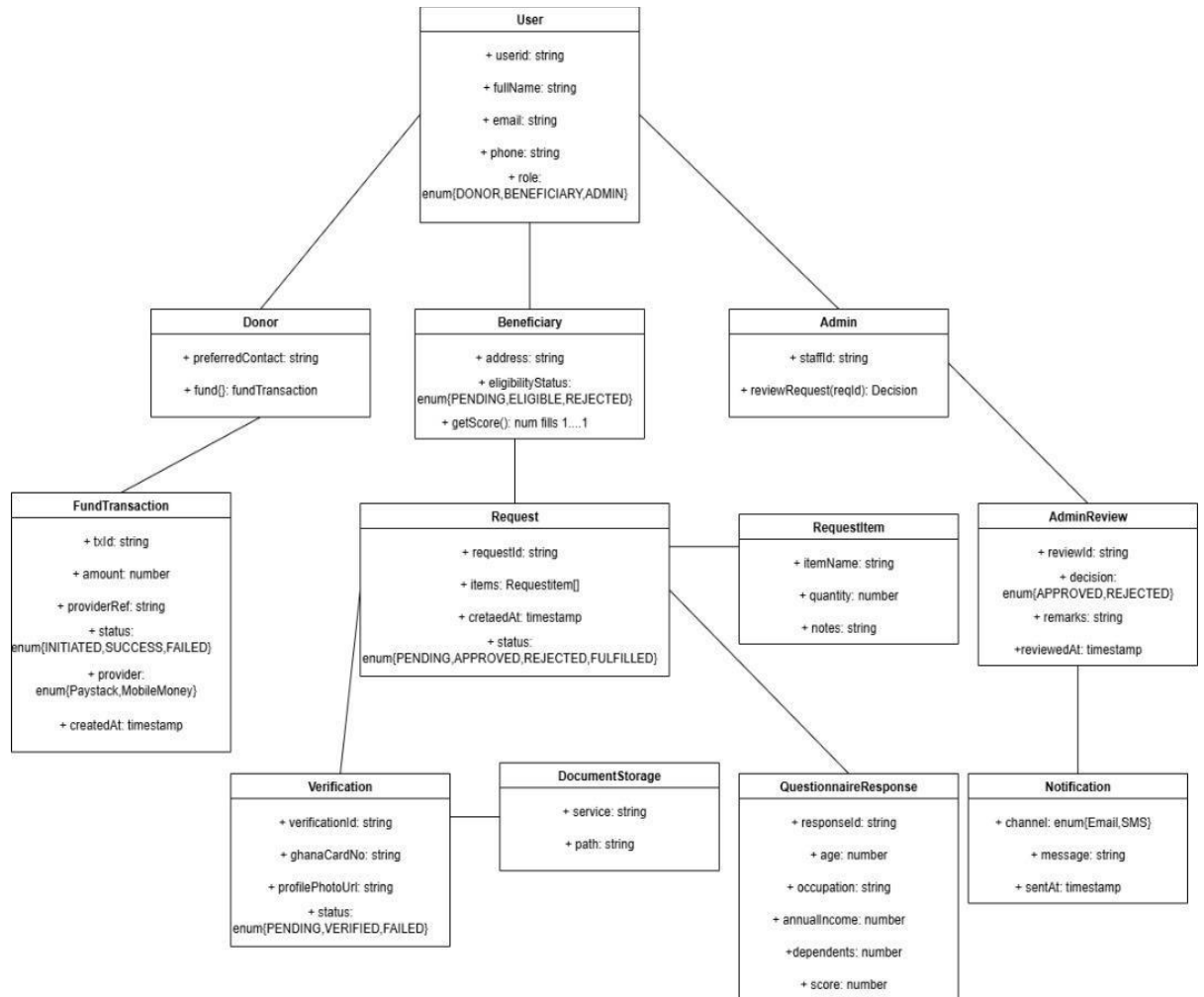
4.8 Requirement Model (Activity Diagram)



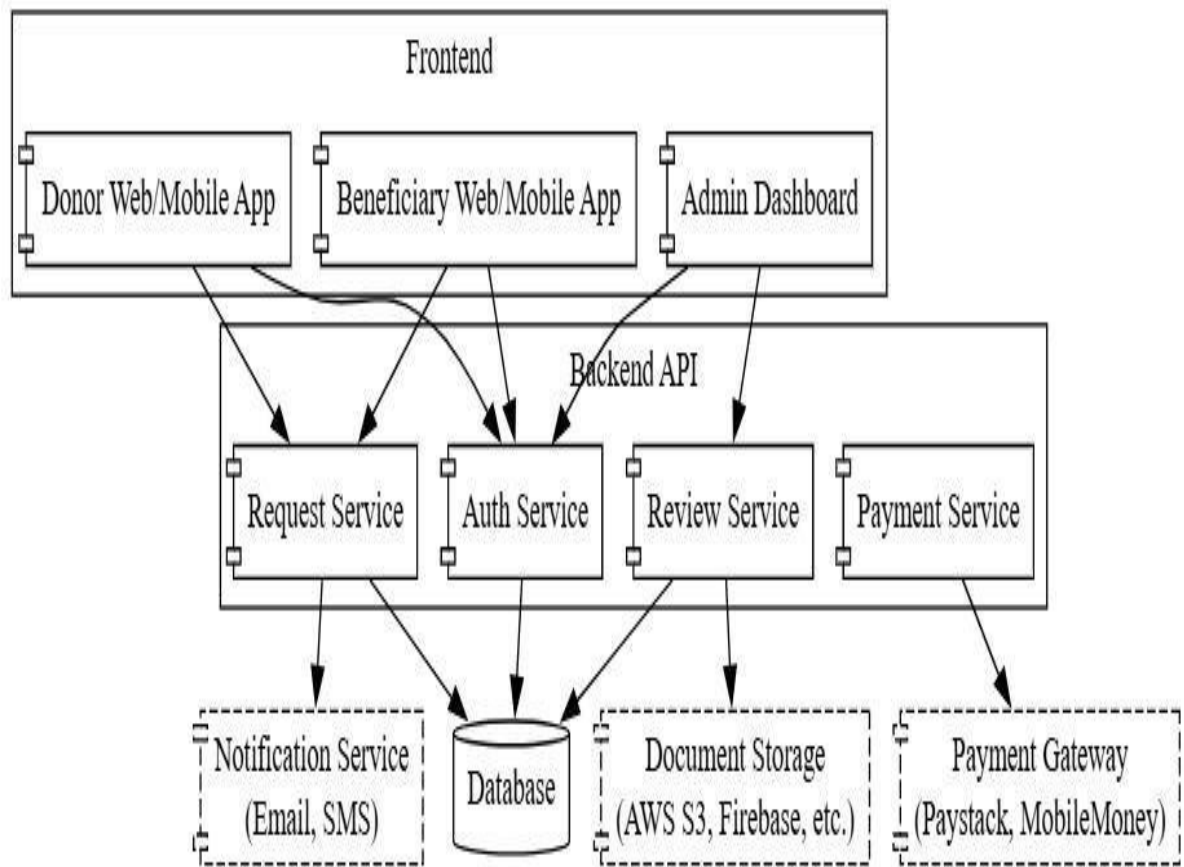
4.9 Requirement Model (Sequence Diagram)



5.0 Requirement Model (Class Diagram)



5.1 Requirement Model (Component Diagram)



CHAPTER V

IMPLEMENTATION AND TESTING

5.2 Development Tools and Platform Consideration

The implementation phase of the AidBridge Web Application requires careful selection of development tools and platforms to ensure maintainability, scalability, and ease of deployment. The chosen tools align with modern web development practices and enable collaborative work among the AidBridge development team.

A. Development Tools

- *Visual Studio Code (VS Code)*: Primary Integrated Development Environment (IDE) used for front-end and back-end development. Extensions such as IntelliSense, integrated terminal, and Git support streamline coding and debugging.
- *Git & GitHub*: Git provides distributed version control, branch management, and version tracking, while GitHub hosts the AidBridge repository for collaboration, issue tracking, and code reviews.
- *Node.js*: JavaScript runtime environment used for building the back-end server. Its event-driven, non-blocking architecture supports sending real-time email notifications via integrated APIs.
- *Express.js*: Web application framework for Node.js, used for routing, middleware, and handling HTTP requests and responses efficiently.
- *Firebase*: Provides authentication, hosting, and a real-time NoSQL database. It stores user credentials, donation records, and beneficiary requests, ensuring instant synchronization across devices.
- *EmailJS and Twilio API*: Integrated for sending email and SMS notifications to beneficiaries and donors regarding approved fundings and requests.

B. Platform Considerations

- *Cross-Platform Compatibility:* Designed with responsive principles using CSS media queries and Bootstrap 5, ensuring accessibility across desktops, tablets, and smartphones.
- *Scalability:* Firebase automatically scales to handle growing numbers of users, requests, and data storage without requiring significant code modifications.
- *Security:* Secure authentication via Firebase, HTTPS connections, and data encryption. Regular security audits ensure the protection of sensitive donor and beneficiary data.

5.3 Testing

Testing is a critical phase of AidBridge development, ensuring performance, security, and usability. Chrome Developer Tools (Inspect Element) was used extensively to debug, optimize, and validate the application throughout development.

Testing Process A) HTML & CSS Testing

- Used the Elements panel to inspect DOM structure and debug layout issues.
- Live-edited CSS to test design changes (colours, spacing, responsiveness).
- Tested media queries for responsiveness on various devices.

B) JavaScript Testing & Debugging

- Console used to test functions and handle errors.
- Breakpoints set in the Sources panel to debug logic step-by-step.
- API calls and form validation logic tested directly in-browser.

C) Network Activity Testing

- Verified Firebase API requests and Email API responses.
- Analyzed request/response times and optimized load speed.
- Simulated weak network conditions to ensure usability in low-connectivity areas.

D) Performance & Memory Testing

- Used Performance tab to check CPU usage and page rendering speed.
- Memory tab used to detect and fix memory leaks.

E) Accessibility & Best Practices

- Lighthouse audits performed for accessibility, SEO, and performance.
- Ensured compliance with WCAG standards for inclusivity.

F) Responsive Design Testing

- Device Toolbar used to simulate smartphones, tablets, and desktops.
- Verified touch interactions like swiping and tapping.

5.4 Unit Testing

Unit testing focused on individual modules (e.g., registration, donation posting). Test cases were executed using browser tools and Firebase's test environment to validate functionality before integration.

Examples:

- Verified "User Registration" correctly validates inputs and stores credentials in Firebase.
- Tested "Donation Posting" to ensure donation details were saved and synced across clients.

5.5 Integration Testing

Integration testing verified that modules (fundings, requests, notifications) work seamlessly together.

Examples:

- A donor submits a fund → System approves → System sends email confirmation via EmailJS.
- A beneficiary requests aid → Request stored in Firebase → Admin approves → Email/SMS with pickup details sent.

5.6 Deployment

The AidBridge Web Application was deployed using Netlify, chosen for its simplicity, scalability, and free hosting plan.

Advantages of Netlify:

- *Ease of Use*: Simple deployment pipeline and intuitive dashboard.
- *Continuous Deployment*: Integrated with GitHub, automatically redeploying when updates are pushed.
- *Free Hosting*: Ideal for student and pilot projects, with generous limits.
- *Custom Domain & SSL*: Enabled secure access (HTTPS) and professional branding.

Deployment Steps:

1. *Prepared Application*: Optimized production build.
2. *Connected GitHub Repository*: Linked source code to Netlify.
3. *Configured Environment Variables*: API keys for Firebase, EmailJS, Twilio API and Paystack set in Netlify.
4. *Deployed Site*: Netlify built and hosted application files.
5. *Custom Domain*: Registered and secured with SSL.
6. *Live Testing*: Verified authentication, donations, Email/SMS notifications, and payment processing.
7. *Monitoring*: Netlify analytics used to track performance and uptime.

CHAPTER VI

ANALYSIS AND DISCUSSION

6.1 Usability Findings

Feedback from pilot users indicated that the user interface was intuitive and easy to navigate. The use of Tailwind CSS ensured a consistent design language, while responsive layouts allowed the system to function smoothly across desktops, tablets, and smartphones. Test users found forms simple to complete, and navigation between the donation and request modules was straightforward.

6.2 Verification and Categorization Insights

Originally, proof-of-need documents were required from beneficiaries; however, this approach was replaced with a structured algorithm-driven questionnaire. The questionnaire included socio-economic and demographic factors such as occupation, household income, age, health condition, and family size.

This method allowed for a fairer and more data-driven categorization of beneficiaries, ensuring that aid could be prioritized for the most vulnerable. Testers and supervisors highlighted this as a more practical and inclusive verification approach. Future integration with the Ghana NIA API was also identified as a valuable extension for automatic Ghana Card validation.

6.3 Request Management Efficiency

The cart-based request process, adapted from e-commerce systems, was effective in allowing beneficiaries to select and request multiple items (e.g., food, clothing, and medical kits) before submission. This streamlined the process, minimized errors, and provided administrators with a clearer overview of beneficiary needs.

6.4 Technical Challenges and Solutions

Several technical challenges were encountered during development:

- *Form Validation:* Designing robust validation for questionnaire inputs.
- *System Security:* Protecting sensitive user data, addressed through Firebase security rules and controlled access.
- *Database Performance:* Optimizing Firebase queries to handle real-time data without latency.
- *Categorization Logic:* Ensuring fair allocation despite diverse donation types.

These were resolved through a combination of secure coding practices, modular database design, and iterative testing.

6.5 Lessons Learned

The iterative development process demonstrated the importance of user-centered design and flexibility in adapting project requirements. In particular, replacing documentary proof-of-need with an algorithm-driven questionnaire-based system was a key learning point, showing how inclusivity and practicality can shape technological solutions.

CHAPTER VII

CONCLUSION AND RECOMMENDATION

7.1 Conclusion

The development of AidBridge has demonstrated that modern web technologies, when applied thoughtfully, can improve aid distribution and social support systems. The project successfully integrated donor management, request processing, and beneficiary verification into a single, user-friendly platform.

Key achievements include:

- Implementing a categorization-based verification system using an algorithm-driven questionnaires.
- Designing a cart-based request module to improve beneficiary experience.
- Enhancing transparency and accountability through secure data handling.

The project not only addresses disaster relief distribution but also shows potential for long-term poverty alleviation and community support programs.

7.2 Recommendations

For continuous improvement and scalability, the following are recommended:

1. *Enhance Categorization with AI/ML:* Train machine learning models to analyze questionnaire data for more accurate and dynamic prioritization of beneficiaries.
2. *Introduce Multilingual and Accessibility Features:* Incorporate local Ghanaian languages and accessibility tools (e.g., screen readers) to broaden inclusivity.
3. *Integrate Geolocation Services:* Use mapping APIs to visualize high-demand areas and optimize distribution.
4. *Develop Partnerships:* Collaborate with NGOs, social welfare agencies, and government bodies to validate beneficiaries and expand outreach.
5. *Expand Donation Categories:* Support not only material donations but also cash transfers, educational resources, and healthcare services.

7.3 Final Remark

With further development and strong partnerships, AidBridge can evolve into a nationally recognized aid distribution platform, bridging the gap between donors and vulnerable populations while ensuring fairness, transparency, and impact.

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APPENDICES

Appendix A: System Features

- Sign-up Page
- Login Page
- Home Page
- About Us Page
- Blog & Educational Resources
- Contact Us Page
- Fund Us Page (with Paystack for donation of funds)
- Request Page, with Ghana Card verification and profile picture upload (beneficiaries specify needed items)
- Algorithm-driven Questionnaire Page (occupation, age, income, health, etc., for verification)
- Notifications System (Email/SMS alert for approvals and updates)
- Admin Dashboard (verify users, approve requests, manage donations)
- Map Integration (locating nearby distribution centres)
- Logout Page

Appendix B: Sample Beneficiary Questionnaire Fields

- Full Name
- Phone Number
- Ghana Card Number (with upload option)
- Occupation
- Age
- Annual Income Bracket (e.g., less than GHS 5,000, GHS 5,000–10,000, above GHS 10,000)
- Health Status (optional but recommended)
- Type of Assistance Needed (food, shelter, clothing, medical, tools)
- Disaster Type (e.g., Flood, Fire, Earthquake)
- Location (Region/District/Community)
- Upload Supporting Evidence (optional – e.g., photo of destroyed property)

Appendix C: System Screenshots

This appendix presents screenshots of the AidBridge web application to illustrate its major features and user interfaces.

1. Sign-up & Login Screens

- *Sign-up Page:* Allows new users (recipients) to register with full details.
- *Login Page:* Secure authentication using Firebase.

AidBridge

Log In [Sign Up](#)

Sign Up

First Name

Last Name

Email Address

Create Password

Register

Get access to thousands of items and relief funds after a disaster.



AidBridge

[Log In](#) [Sign Up](#)

Email

Password

Log In

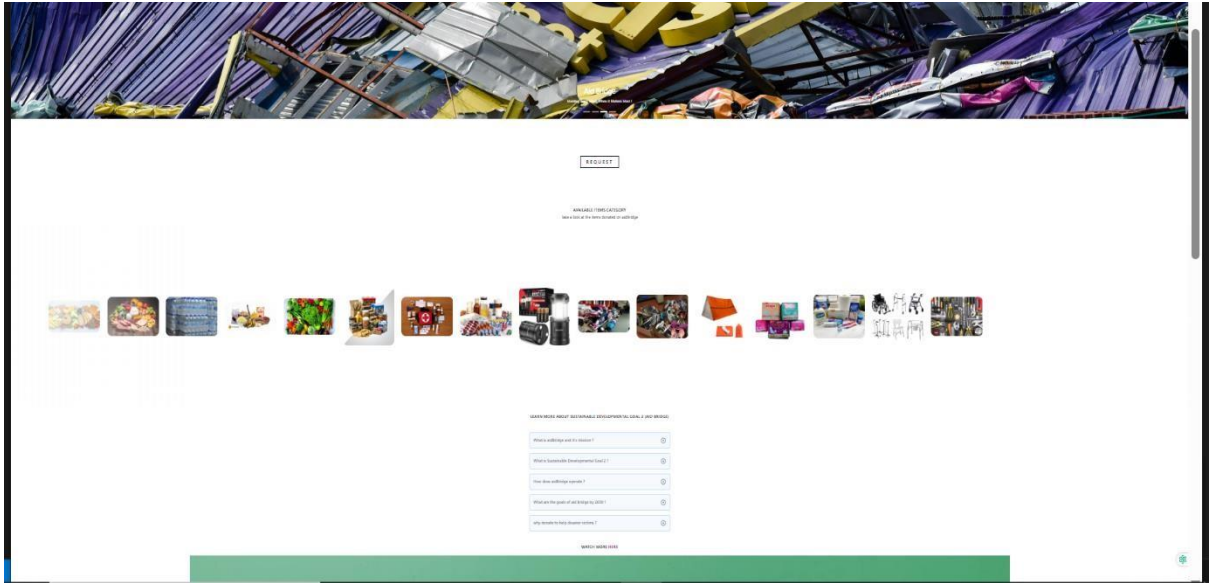
[Forgot password](#)

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2. Home Page & Navigation

- *Home Page:* Overview of AidBridge with quick links to donations, requests, and blogs.
- *Navigation Bar:* Provides easy access to system modules.



3. Ghana card and profile verification Page

- *Identity verification:* Verifies the users' ID cards and profile pictures to see if they match.

Ghana ID Card & Face Verification System

Step 2: Upload Your Selfie and Ghana ID Card

Manually entered ID: [redacted]

Upload Selfie Photo: [Choose File] (Photo: selfie.jpg)

Upload Ghana ID Card: [Choose File] (ID: 202403-4732_487.jpg)

[Start Verification] [Cancel]

Verification Results:

- ID VERIFICATION PASSED
Provided ID number matches
- FACE VERIFICATION PASSED
Facial features match
- IDENTITY VERIFICATION SUCCESSFUL
Self ID and face verification passed

Verification Details:

Manually Entered ID: [redacted]

ID from Card is: [redacted]

ID Match Status: MATCH

Face Match Status: 100%

Face Similarity Score: 100%

Verification Date: 31/10/2024, 1:03:24 PM

[Return to Dashboard]

4. Donation & Request Pages

- *Fund Us Form:* Donors donate funds to the platform with the help of Paystack.
- *Request Form:* Beneficiaries request assistance.

Make a Payment

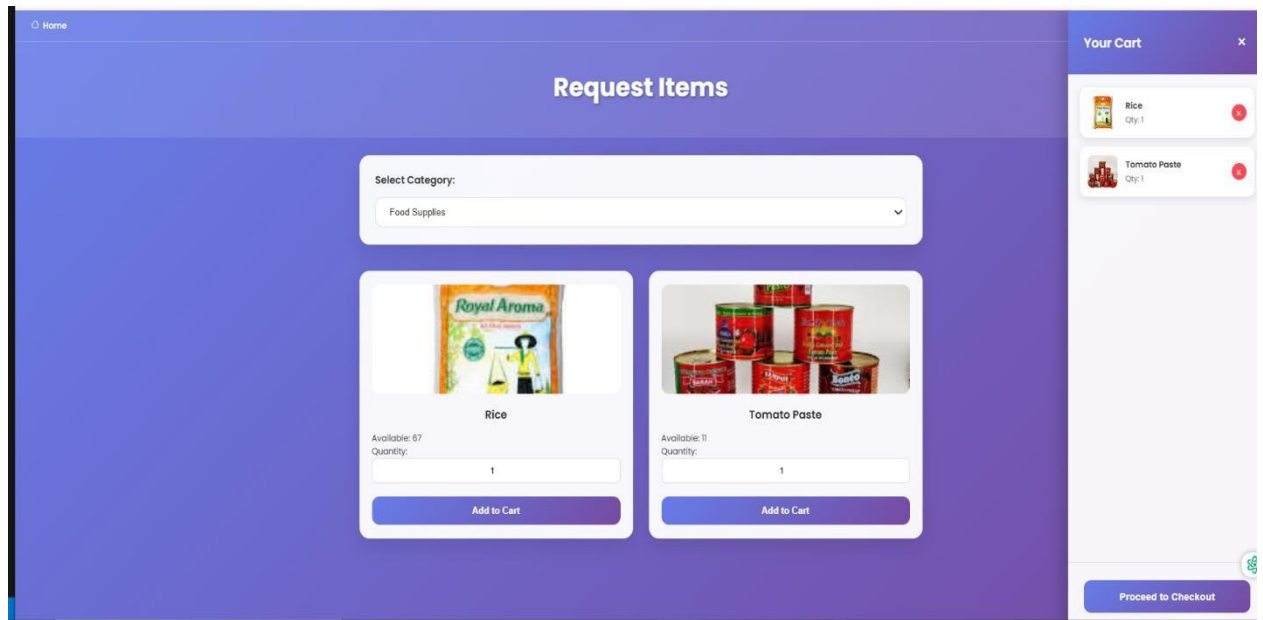
Email Address

Amount

First Name

Last Name

Pay Now

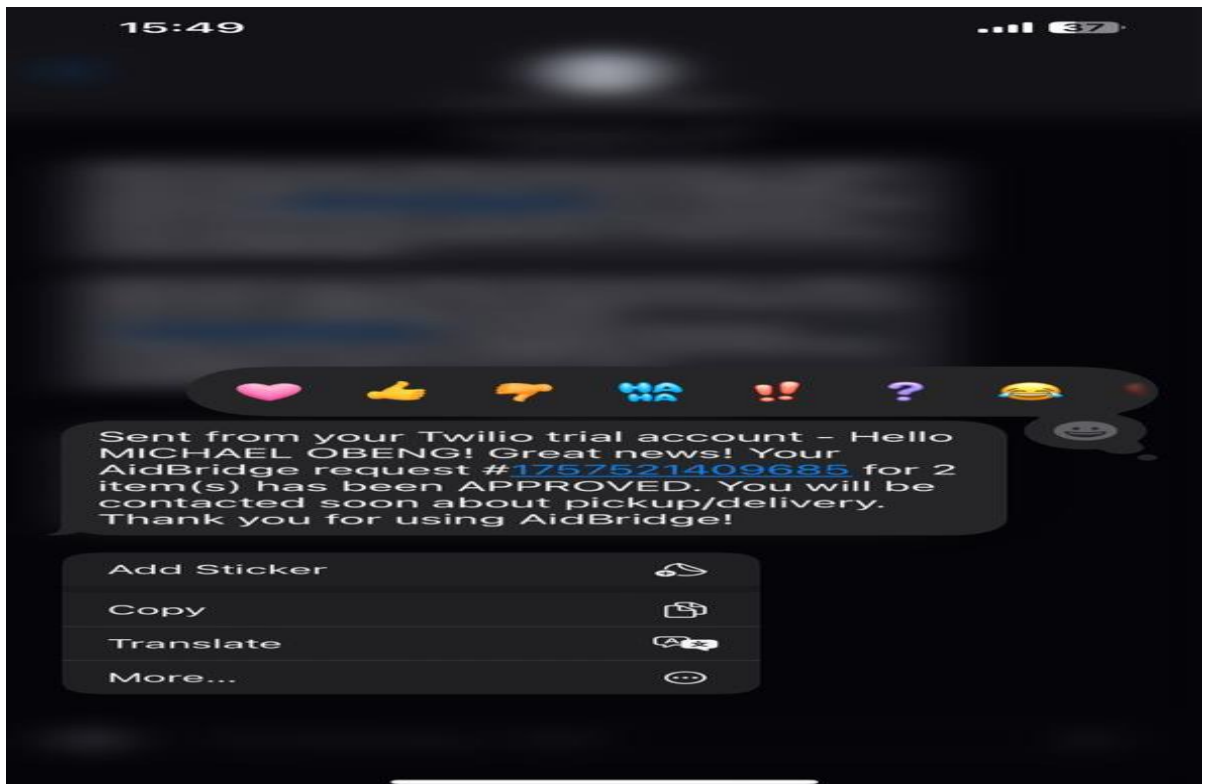
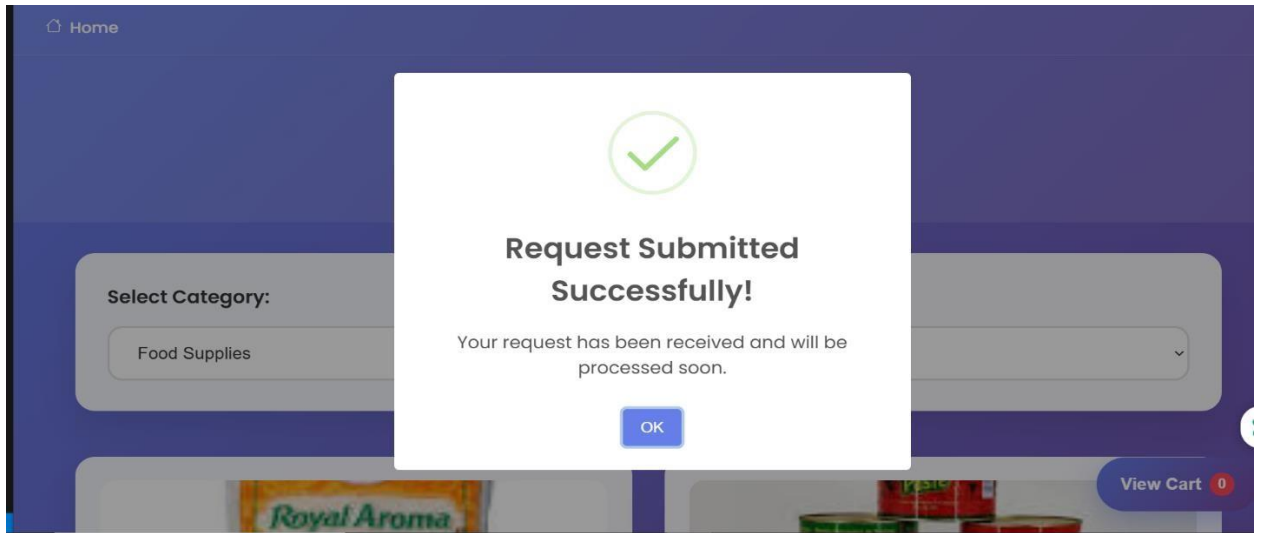


5. Beneficiary Questionnaire

- *Algorithm-driven Questionnaire Page:* Collects socio-economic and demographic data (occupation, income, age, etc.) to categorize need levels.

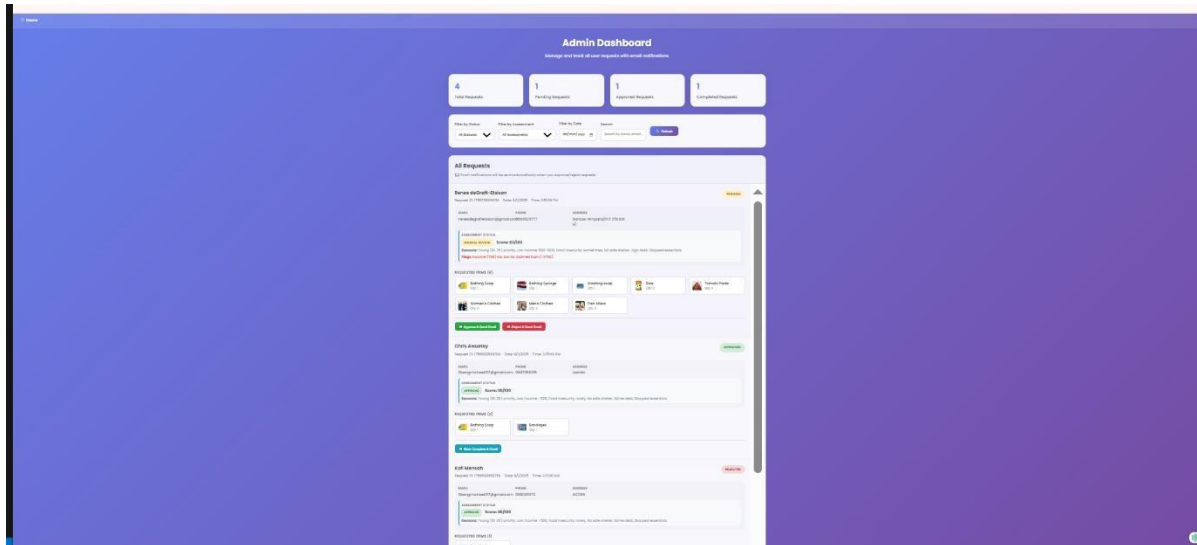
6. Notifications/Email,SMS

- *Confirmation Messages:* Donors/recipients receive an email, SMS or on-screen confirmation once actions are approved.



7. Admin Dashboard

- *Dashboard Overview:* Admin verifies Ghana Card details, approves requests, and confirms donations.
- *Inventory Management:* Admins manage inventories received from donors.



AidBridge

Back

Add Item to Inventory

Item Name:

e.g. Rice

Image URL:

https://example.com/image.jpg

Quantity:

Category:

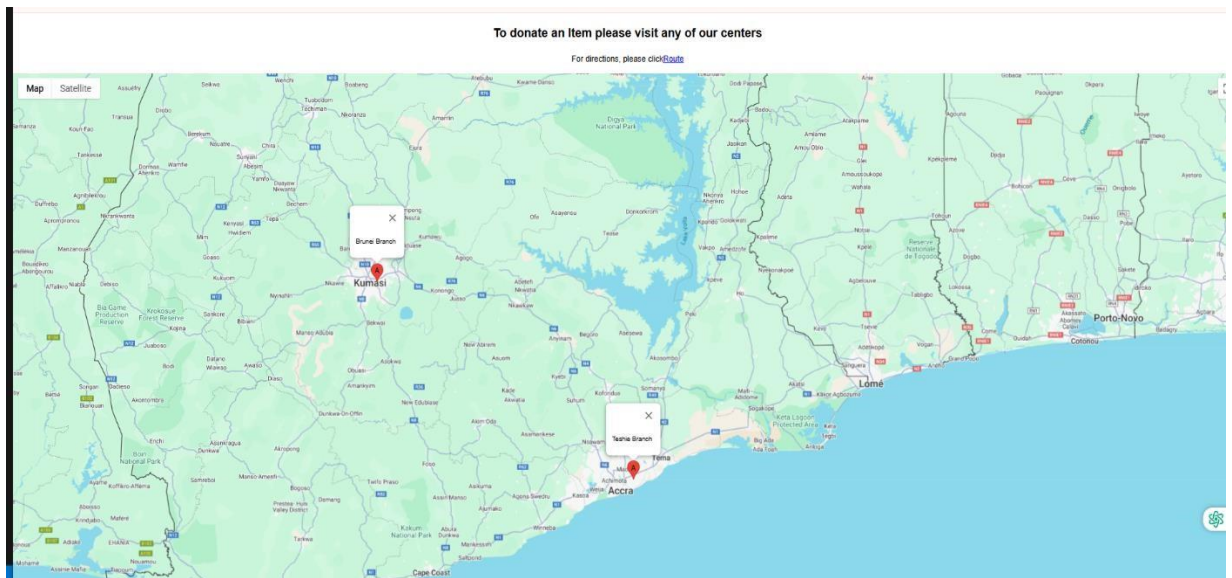
--Select Category--

Add Item

An illustration of a man with a beard, wearing a green t-shirt, sitting at a table and eating from a white bowl filled with green vegetables and brown grains. A small potted plant with green leaves stands to his left. Above him is a circular light green background. To the right of the man is a black speech bubble containing a green circle with a diagonal slash over it, indicating a prohibition or warning.

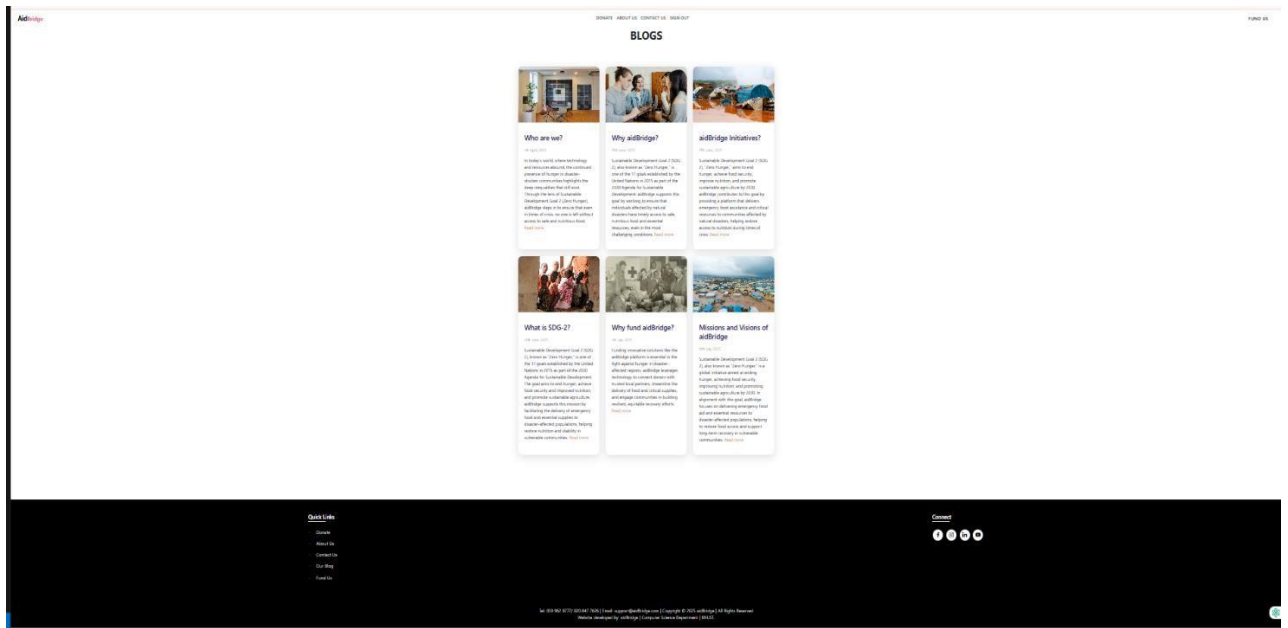
8. Map Integration

- *Map View:* Shows nearby food bank centres and donation distribution points using Google Maps.



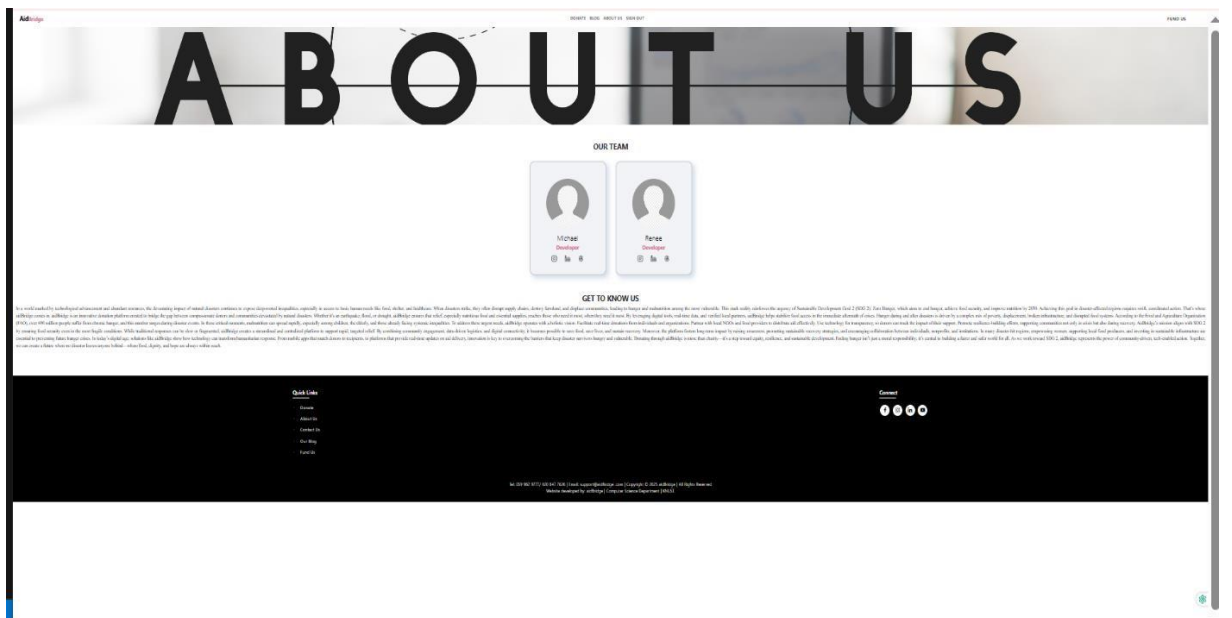
9. Blog & Educational Resources

- *Blog Page:* Provides educational content on disaster preparedness, food security, and health tips.



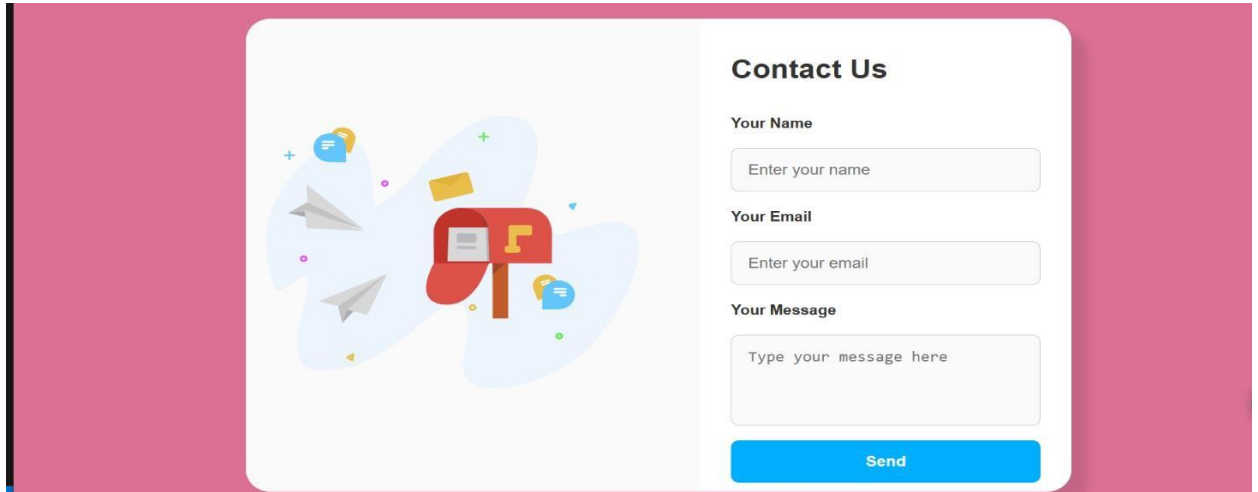
10. About-Us Page

- *About-us*: Provides information about aidBridge and its team members.



11. Contact Us

- *Contact page:* Donors and recipients can contact the team about anything they don't understand or need assistance on.

A screenshot of a 'Contact Us' form. On the left is a light blue illustration of a mailbox with a red flag, surrounded by paper airplanes, a lightbulb, and a plus sign. On the right is a white form titled 'Contact Us'. It contains three input fields: 'Your Name' with placeholder text 'Enter your name', 'Your Email' with placeholder text 'Enter your email', and 'Your Message' with placeholder text 'Type your message here'. Below these fields is a blue 'Send' button.

Contact Us

Your Name
Enter your name

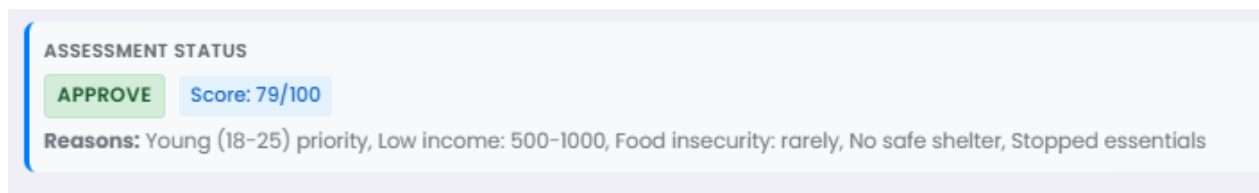
Your Email
Enter your email

Your Message
Type your message here

Send

12. Algorithm-Driven Questionnaire

- Scores the beneficiary based on the structure of the algorithm and the answers the beneficiary provides.

A screenshot of an 'ASSESSMENT STATUS' section. It features a green 'APPROVE' button and a blue 'Score: 79/100' label. Below these is a line of text listing reasons for the assessment: 'Reasons: Young (18-25) priority, Low income: 500-1000, Food insecurity: rarely, No safe shelter, Stopped essentials'.

ASSESSMENT STATUS

APPROVE Score: 79/100

Reasons: Young (18-25) priority, Low income: 500-1000, Food insecurity: rarely, No safe shelter, Stopped essentials

Appendix D: System Functionality Source Code

// Income range mapping for calculations

```
const incomeRangeMapping = {  
  "<500": 400,  
  "500-1000": 750,  
  "1001-2000": 1500,  
  ">2000": 3000  
};
```

// Scoring algorithm

```
function computeScore(data) {  
  let score = 0;  
  const reasons = [];  
  const flags = [];
```

// Age priority

```
  if (data.ageRange === "18-25") {  
    score += 15;  
    reasons.push("Young (18-25) priority");  
  }  
  else if (data.ageRange === "60+") {  
    score += 15;  
    reasons.push("Senior (60+) priority");  
  }  
  else if (data.ageRange === "26-30") {  
    score += 8;  
    reasons.push("Medium priority age (26-30)");  
  }  
  else if (data.ageRange === "31-59") {  
    score += 5;  
    reasons.push("Lower priority age (31-59)");  
  }
```

// Income

```
  const incomeWeights = {"<500":25,"500-1000":18,"1001-2000":8,">2000":0};  
  score += incomeWeights[data.incomeRange] || 0;  
  if (incomeWeights[data.incomeRange]) reasons.push("Low income: " +  
data.incomeRange);
```

// Meals

```
  const mealWeights = {never:25, rarely:18, sometimes:10, always:0};  
  score += mealWeights[data.mealFrequency] || 0;  
  if (mealWeights[data.mealFrequency] && data.mealFrequency !== 'always') {  
    reasons.push("Food insecurity: " + data.mealFrequency);  
  }
```

// Shelter

```
  if (data.hasSafeShelter === "no") {
```

```

    score += 18;
    reasons.push("No safe shelter");
}

```

// **Loans**

```

if (data.hasLoans === "yes") {
    const loan = Number(data.loanAmount) || 0;
    if (loan > 100000) {
        flags.push("Loan amount exceeds limit (100000)");
    } else {
        if (loan >= 5000) {
            score += 12;
            reasons.push("High debt");
        }
        else if (loan >= 2000) {
            score += 8;
            reasons.push("Medium debt");
        }
        else if (loan > 0) {
            score += 4;
            reasons.push("Some debt");
        }
    }

    const incomeVal = incomeRangeMapping[data.incomeRange];
    if (incomeVal && loan > incomeVal * 5) {
        flags.push("Income (" + incomeVal + ") too low for claimed loan  
(>" + (incomeVal * 5) + ")");
    }
}
}

```

// **Essential services**

```

if (data.essentialStopped === "yes") {
    score += 10;
    reasons.push("Stopped essentials");
}

```

// **Inconsistencies**

```

    if (data.incomeRange === ">2000" && (data.mealFrequency === "rarely" ||
data.mealFrequency === "never")) {

```



```

    flags.push("High income but food insecure");
  }

  // Household size flag
  if (data.householdSize && Number(data.householdSize) > 10) {
    flags.push("Unusually large household size (>10)");
  }

  // Occupation vs Income consistency check

  const occupationIncomeRanges = {
    public: [418, 33855],
    private: [2500, 5000],
    informal: [1200, 2200],
    selfemployed: [1200, 7000],
    unskilled: [500, 1690],
    skilled: [2000, 10000],
    student: [0, 1000],
    unemployed: [0, 500]
  };

  const occ = data.occupation;
  const incomeVal = incomeRangeMapping[data.incomeRange];
  if (occ && incomeVal && occupationIncomeRanges[occ]) {
    const [min, max] = occupationIncomeRanges[occ];
    if (incomeVal < min*0.8 || incomeVal > max*1.2) {
      flags.push("Income inconsistent with occupation: " + occ + " at GHS " +
incomeVal);
    }
  }

  // Cap score to 100

  if (score > 100) score = 100;

  // Status

  let status = "Manual Review";
  if (score >= 70 && flags.length === 0) status = "Approve";
  else if (score < 35) status = "Reject";

  return {score, reasons, flags, status};
}

```